

Padala Hemanth Subbi Reddy

ML Engineer — Computer Vision — Deep Learning

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Profile

ML Engineer with production experience in computer vision, GPU inference optimization, and real-time video analytics. Experienced in PyTorch, TensorRT, CUDA, object detection, tracking, and production ML systems. Built and optimized large-scale CV inference pipelines, including a 1,603 FPS TensorRT engine and a 14,370 FPS multi-camera analytics pipeline on 8x NVIDIA H100 GPUs. Incoming M.Sc. Data Science student at RWTH Aachen University, Germany.

Education

RWTH Aachen University

M.Sc. Data Science

Oct 2026 – 2028 (Expected)
Aachen, Germany

Vellore Institute of Technology (VIT)

B.Tech in Electronics and Computer Engineering

June 2026
CGPA: 8.82/10

Technical Skills

Machine Learning: PyTorch, TensorFlow, HuggingFace, YOLOv8, Object Detection, Tracking, Model Fine-tuning, Quantization, RAG, LLMs

Computer Vision: OpenCV, CUDA, TensorRT, Video Analytics, NMS, Multi-camera Analytics, Incident Detection, Image/Video Processing

LLM / AI: LangChain, LangGraph, FAISS, ChromaDB, Llama, Mistral, Whisper, ASR, Agentic Workflows

Engineering: Python, C/C++, FastAPI, REST APIs, Docker, Redis, PostgreSQL, AWS, Git, Linux, SQL

Experience

Matrice AI

ML Engineer Intern

Apr 2026 – Present
San Francisco, USA (Remote)

- Diagnosed a production weapon-detection false-positive failure mode on low-quality CCTV footage; mined hard negatives and fine-tuned only the classification head, raising F1 from **78% to 85%** on low-resolution feeds while preserving localization behavior and deployment compatibility.
- Wrote custom CUDA kernels for GPU-resident video preprocessing and post-processing, including NV12→RGB conversion, bilinear/letterbox resize, zero-copy crops, and fused GPU NMS operations.
- Built a custom TensorRT inference engine achieving **1,603 FPS vs. 160 FPS** with stock Ultralytics inference; increased camera-stream service rate from **26% to 100%** and reduced p50 latency from **865ms to 248ms**.
- Implemented INT8 quantization with **45% lower per-batch execution time** (67ms→37.1ms) and **41% smaller engine footprint**, validating accuracy parity against the FP16 baseline.
- Optimized a production video-analytics pipeline to sustain **14,370 FPS across 1,000 simulated camera streams** on 8x NVIDIA H100 GPUs at 97% utilization; eliminated synchronization and locking bottlenecks for a 2.6x throughput gain.
- Unified weapon, violence, and loitering incident systems into a shared severity-state-machine architecture and owned analytics-to-Redis publishing across 8+ CV applications; shipped **8 production CV applications end-to-end** for release.

F-AI

AI Engineer Intern — GenAI & LLM Systems

Jan 2026 – Mar 2026
Bengaluru, India (Remote)

- Built a production RAG system for financial Q&A using semantic chunking, embeddings, retrieval, and LLM generation; evaluated RAG versus fine-tuning trade-offs and developed grounding/evaluation workflows.
- Reduced LLM inference latency by nearly **3x** through structured prompting, sliding-window context, and asynchronous LangGraph orchestration; deployed Llama and Mistral locally.

Samsung PRISM

Student Researcher

Sept 2024 – June 2025
Bengaluru, India (Remote)

- Designed an ECA-enhanced U-Net for speech super-resolution, achieving a **10.2% relative WER reduction** on LibriSpeech for downstream ASR.
- Benchmarked streaming architectures including SEANet under real-time causal and non-causal latency-accuracy trade-offs.

Projects

Whisper ASR Fine-Tuning for Dhivehi

GitHub

- Fine-tuned OpenAI Whisper on Common Voice 13.0 for low-resource Dhivehi ASR and evaluated WER/CER across training and validation splits under noisy audio conditions.
- Analyzed cross-attention head behavior associated with transcription hallucinations, connecting model-level observations to speech-enhancement experiments conducted during Samsung PRISM research.

RAG Agent on Federal Register

GitHub

- Built an end-to-end Llama2 RAG system over U.S. Federal Register documents using semantic chunking, FAISS/ChromaDB retrieval, and source-grounded generation; achieved **92% retrieval accuracy**.
- Deployed the system with asynchronous FastAPI and Docker, enabling low-latency document retrieval and conversational Q&A through a full-stack interface.

Research & Achievements

- **Amazon ML Challenge 2025:** Ranked **94 / 7,000+ teams (Top 1.5%)** with a multimodal ML pipeline.
- **4 peer-reviewed publications, 13+ citations** in computer vision, remote sensing, and deep learning. [Google Scholar](#)
- **IEEE CICT 2023:** YOLOv8-based weed and disease detection in drone imagery (**12 citations**).
- **IEEE Access 2025:** SPSSA-optimized Random Forest for remote-sensing change detection.
- **IEEE Access 2026:** Phase Congruency-enhanced YOLO for real-time pothole detection on edge devices.
- **International Journal of Remote Sensing 2025:** Modified U-Net for waterbody change detection in optical satellite imagery.